MoDPA: Investigating diseases through Modification-Dependent Protein Associations

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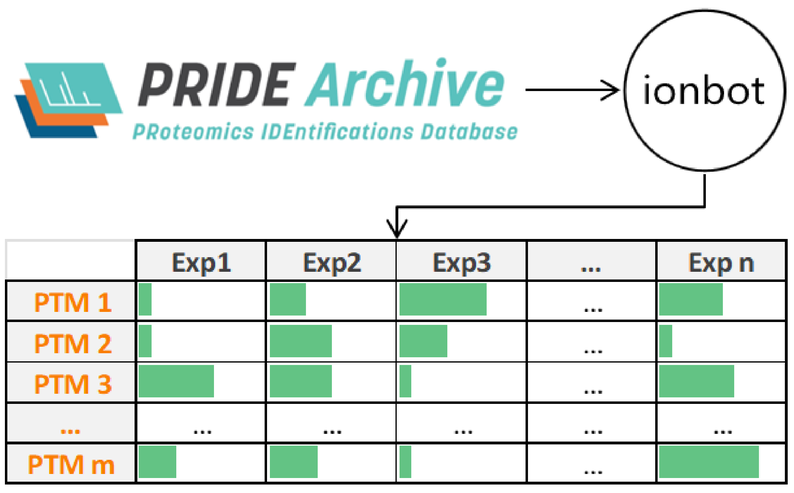
Protein-Protein Interactions **(PPIs)** and Post-Translational Modifications **(PTMs)** are vital in regulating biological processes[1,2], but their interplay is still poorly understood. To address this, I am developing a computational approach called **MoDPA** (Modification-Dependent Protein Associations), to identify co-occurring PTMs.

1. PTM identification and quantification

Raw data files were retrieved from **PRIDE** and reprocessed with **ionbot** in **open modification search** mode.

PTMs were quantified by normalising the counts of identified peptides carrying a modification by the coverage of the modified residue.

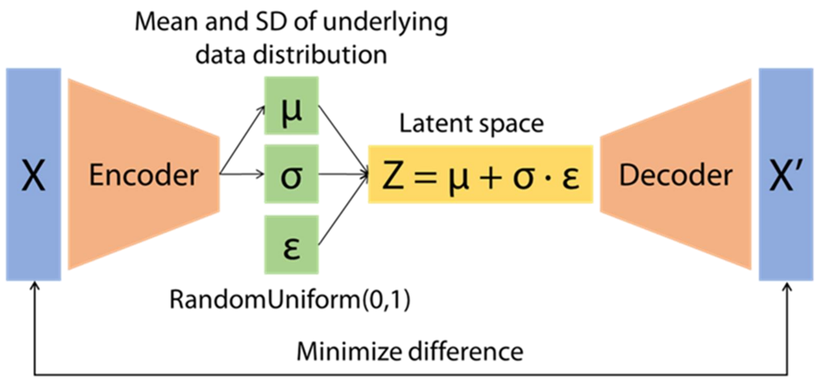
$$\frac{\#Peptides_{modified\ on\ site\ S}}{\#Peptides_{spanning\ site\ S}}$$



2. Dimensionality reduction with VAE

Variational Autoencoders (VAEs) are deep-learning models that, given a multi-dimensional vector *X*, compress it to a low-dimensional latent space *Z*, then decompress *Z* back into *X*[3,4].

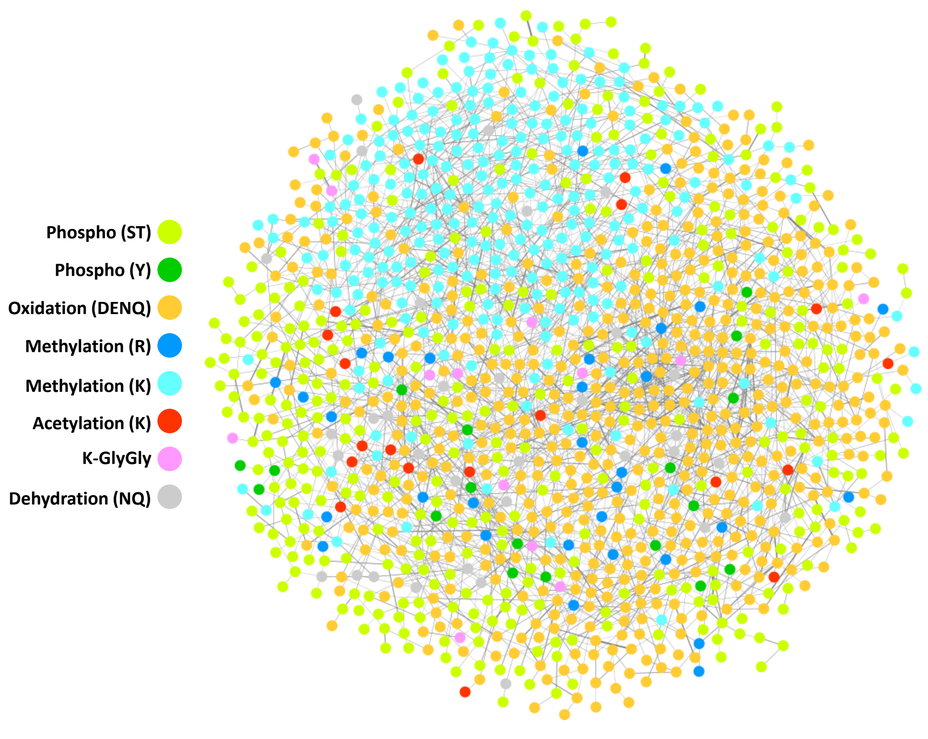
By training a VAE on the high-dimensional PTM data, I obtained a low-dimensional representation of the data for PTMs correlation analysis.

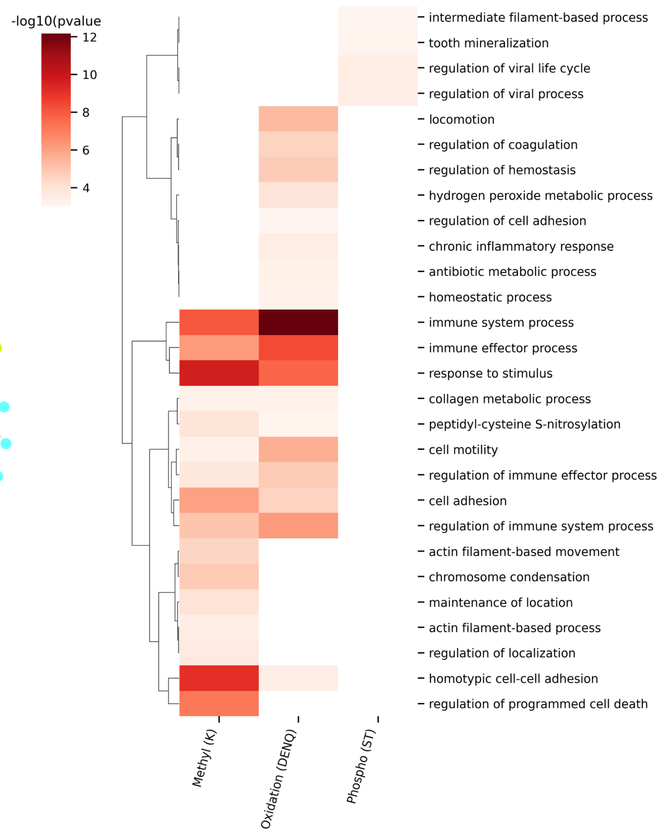


3. PTMs associations network

I used correlations calculated from the latent space to build a weighted network of PTM associations.

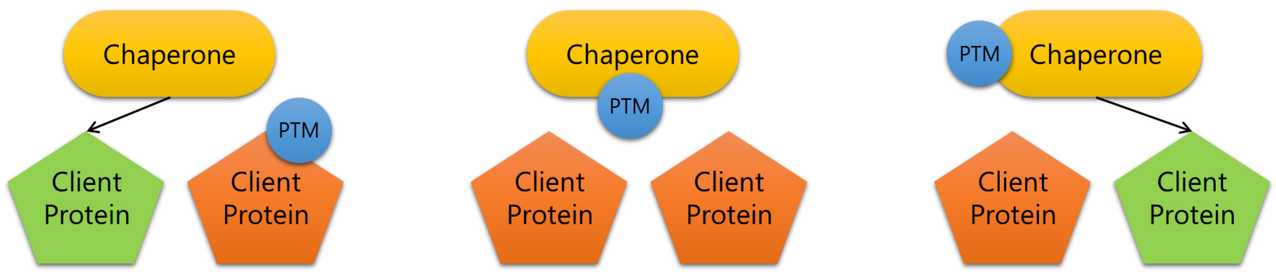
Preliminary network analysis identified clusters of interconnected PTMs on proteins associated with **immunity**, with different PTMs playing different roles within the context of immune response.





4. Case study: Molecular Chaperones

I plan to apply my approach to study how PTMs affect the interactions between **molecular chaperones** and their client proteins, and how they affect **proteostasis**[5].



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